

PRE-BOARD EXAMINATION (2024-25)

CLASS-X

SUBJECT- SCIENCE (086)

Max. Marks: 80

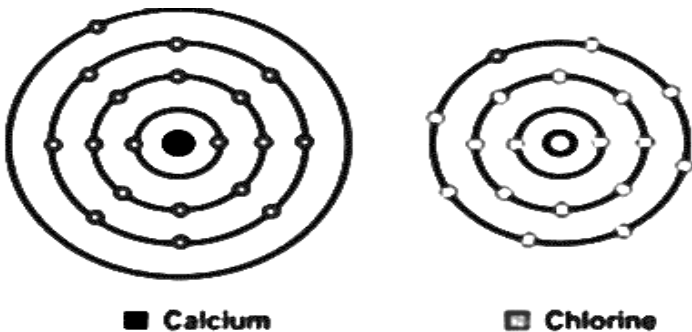
Time Allowed: 3 hours

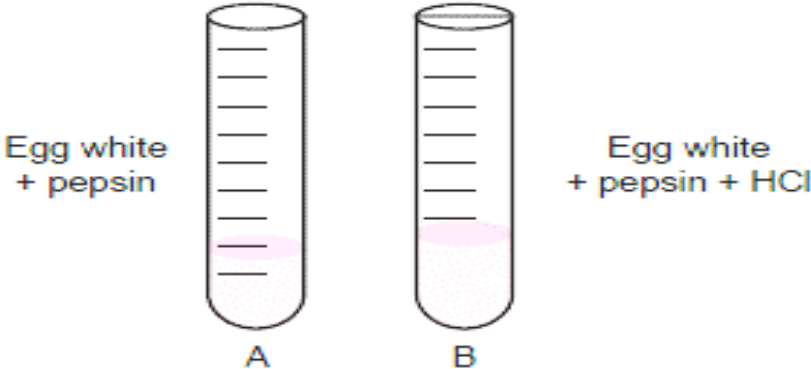
General Instructions:

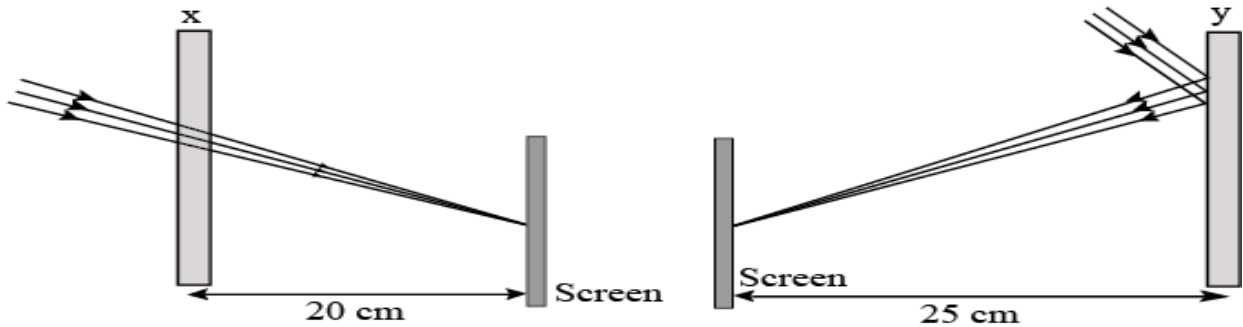
- All questions would be compulsory. However, an internal choice of approximately 33% would be provided. 50% marks are to be allotted to competency-based questions.
- Section A would have 16 simple/complex MCQs and 04 Assertion-Reasoning type questions carrying 1 mark each.
- Section B would have 6 Short Answer (SA) type questions carrying 02 marks each.
- Section C would have 7 Short Answer (SA) type questions carrying 03 marks each.
- Section D would have 3 Long Answer (LA) type questions carrying 05 marks each.
- Section E would have 3 source based/case based/passage based/integrated units of assessment (04 marks each) with sub-parts of the values of 1 or 2 marks.

Section-A

Question 1 to 16 are multiple choice questions. Only one of the choices is correct. Select and write the correct choice as well as the answer to these questions.

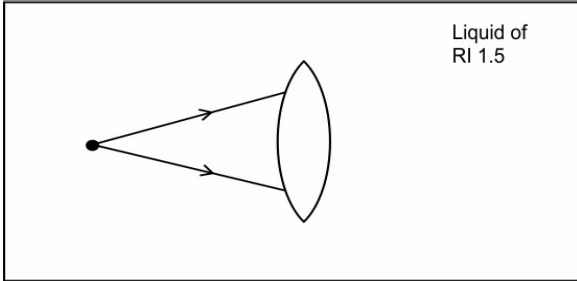
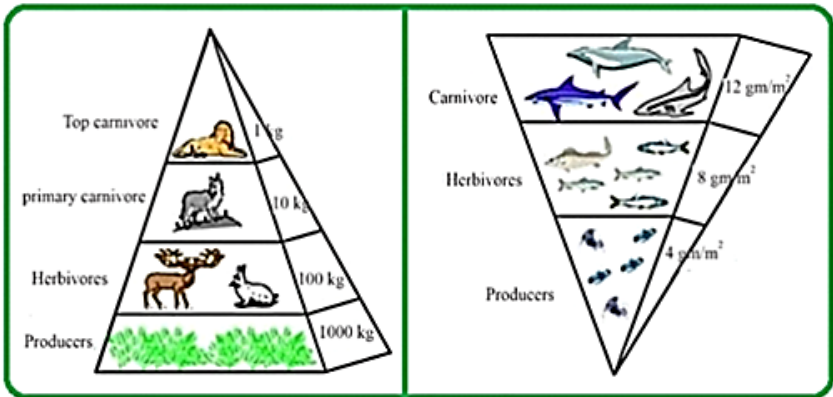
1	Which of the following observations is correct for the above given experimental diagram?  A. Red-coloured copper is reduced to black-coloured copper(I) oxide. B. Red-coloured copper is oxidized to red-coloured copper(I) oxide. C. Red-coloured copper is reduced to black-coloured copper (II) oxide. D. Red-coloured copper is oxidized to black coloured copper(II) oxide.	1
2	A scientist is attempting to represent an ionic bond between calcium and chlorine. The figure below shows the progress he has made so far.  What should be the next step in his representation of the ionic bond? A. Transfer an electron from the calcium atom to the chlorine atom. B. Transfer an electron from the chlorine atom to the calcium atom. C. Add another chlorine atom to accept an electron from the calcium atom. D. Add another calcium atom to donate an electron to the chlorine atom.	1
3	The aqueous solution of one of the following salts will turn red litmus to blue. This salt is: A. potassium sulphate B. sodium carbonate C. ammonium sulphate D. potassium carbonate	1
4	Which of the following phenomena occur when a small amount of acid is added to water? A. Dilution B. Ionization C. Neutralization D. Both (A) and (B)	1

5	Consider the following equation of the chemical reaction of a metal M : $4M + 3O_2 \rightarrow 2M_2O_3$ This equation represents: A. combination reaction as well as reduction reaction. B. combination reaction as well as oxidation reaction. C. decomposition reaction as well as oxidation reaction. D. oxidation reaction as well as displacement reaction.	1
6	If the number of proton in one atom of an element X is 20, then the number of electron in its ion X^{+2} will be: A. 20 B. 19 C. 18 D. 16	1
7	A white precipitate will be formed by adding dilute sulphuric acid to: A. $CuSO_4$ solution B. NaCl solution C. $BaCl_2$ solution D. Na_2SO_4 solution	1
8	The correct sequence of anaerobic reactions in yeast is: A. Glucose $\rightarrow$ (cytoplasm) Pyruvate $\rightarrow$ (mitochondria) Ethanol + Carbon dioxide B. Glucose $\rightarrow$ (cytoplasm) Pyruvate $\rightarrow$ (cytoplasm) Lactic acid C. Glucose $\rightarrow$ (cytoplasm) Pyruvate (mitochondria) $\rightarrow$ Lactic acid D. Glucose $\rightarrow$ (cytoplasm) Pyruvate $\rightarrow$ (cytoplasm) Ethanol + Carbon dioxide	1
9	Which one of the following statement is correct about the human circulatory system? A. Blood transports only oxygen and not carbon dioxide. B. Human heart has five chambers. C. Valves ensure that the blood does not flow backwards. D. Both oxygen-rich and oxygen-deficient blood gets mixed in the heart.	1
10	A student sets up an experiment to study the role of enzymes in digestion of food. Digestion of protein will take place in:  The diagram shows two test tubes, A and B, each with horizontal markings. Test tube A is labeled 'Egg white + pepsin' and contains a pinkish liquid. Test tube B is labeled 'Egg white + pepsin + HCl' and contains a more opaque, white substance. A. test tube A as pepsin will breakdown complex molecules into simple molecules. B. test tube B as HCl will breakdown protein into simple molecules. C. test tubes A as pepsin will breakdown protein into simple molecules. D. test tube B as HCl will activate pepsin for breakdown of protein into simple molecules.	1
11	In a synapse, a chemical signal is transmitted from: A. dendritic end of one neuron to axonal end of another neuron. B. axon to the cell body of the same neuron. C. cell body to axonal end of the same neuron. D. axonal end of one neuron to dendritic end of another neuron.	1
12	The two organisms that can regenerate fully from their cut body parts are : A. hydra and planaria B. paramecium and hydra C. hydra and amoeba D. planaria and leishmania	1
13	An object is placed in front of a convex mirror. Its image is formed: A. at a distance equal to the object distance in front of the mirror. B. at twice the distance of the object in front of the mirror. C. half the distance of the object in front of the mirror. D. behind the mirror and it's position varies according to the object distance.	1

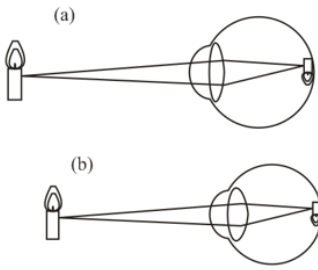
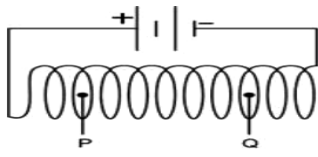
14	 Study the given ray diagrams and select the correct statement from the following. A. Device X is a concave mirror and device Y is a convex lens, whose focal lengths are 20cm and 25cm respectively. B. Device X is a convex lens and device Y is a concave mirror, whose focal lengths are 10cm and 25cm respectively. C. Device X is a concave lens and device Y is a convex mirror, whose focal lengths are 20cm and 25cm respectively. D. Device X is a convex lens and device Y is a concave mirror, whose focal lengths are 20cm and 25cm respectively.	1
15	Ozone forms by combination of free oxygen atoms and oxygen molecules. How do free oxygen atoms form at higher levels of atmosphere? A. By splitting of molecular oxygen into free oxygen atoms in the presence of low energy UV radiations. B. By splitting of a molecular oxygen into free oxygen atoms in presence of high energy UV radiations. C. By the combination of two molecular oxygen in the presence of high energy UV radiations. D. By the combination of two free oxygen atoms in the presence of lower energy UV radiations	1
16	Accumulation of non-biodegradable pesticides in the food chain in increasing amount at each higher trophic level is known as A. eutrophication B. pollution C. biomagnification D. accumulation	1
Question No. 17 to 20 consist of two statements – Assertion (A) and Reason (R). Answer these questions by selecting the appropriate option given below: A. Both A and R are true, and R is the correct explanation of A. B. Both A and R are true, and R is not the correct explanation of A. C. A is true but R is false. D. A is false but R is true		
17	Assertion (A): The extraction of metals from their sulphide ores cannot take place without roasting of the ore. Reason (R): Roasting converts sulphide ores directly into metals.	1
18	Assertion (A): A convex mirror always forms an image behind it and the image formed is virtual. Reason (R): According to the sign convention, the focal length of a convex mirror is positive.	1
19	Assertion (A): Variations always provide a survival advantage to an organism. Reason (R): Variations can be caused due to incorrect DNA copying.	1
20	Assertion (A): Decomposers act as cleaning agents of the environment. Reason (R): The decomposers recycle waste material in the hydrosphere.	1

Section-B

Question No. 21 to 26 are very short answer questions

21	Differentiate between alveoli and nephron on the basis of the following points: <table><tr><th>S. No.</th><th>Feature</th><th>Alveoli</th><th>Nephron</th></tr><tr><td>1</td><td>Structure and location</td><td></td><td></td></tr><tr><td>2</td><td>Function</td><td></td><td></td></tr></table>	S. No.	Feature	Alveoli	Nephron	1	Structure and location			2	Function			2
S. No.	Feature	Alveoli	Nephron											
1	Structure and location													
2	Function													
22	Balance the following equations: A. $\text{NaOH} + \text{H}_2\text{SO}_4 \rightarrow \text{Na}_2\text{SO}_4 + \text{H}_2\text{O}$ B. $\text{Mg}(\text{OH})_2 + \text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2\text{O}$	2												
23	<u>Attempt either option A or B.</u> A. If a plant is releasing carbon dioxide and taking oxygen during the day does it means that there is no photosynthesis occurring? Justify your answer. OR B. Give the name of the enzyme present in the saliva in the buccal cavity. State the gland which produces it and what would happen to the digestion process due to this secretion?	2												
24	A lens made of material with refractive index 1.5 is immersed in a liquid with refractive index 1.5. The diagram below shows two rays incident on the lens when it is immersed in the liquid.  Copy the diagram and draw the light rays after they pass through the lens. Justify your diagram.	2												
25	Shown below are two food pyramids.  The pyramid representing the land ecosystem is traditional with producers being greater in mass than primary consumers and so on. Sometimes, in aquatic ecosystems, an inverted pyramid exists. Here, the total mass of producers (phytoplankton) is much smaller than the top consumers (big fishes). A. Which level is likely to have the most amount of energy in such an aquatic ecosystem? Give a reason to support your answer. B. Such aquatic ecosystems are not considered to be sustainable. Justify this statement.	2												
26	<u>Attempt either option A or B.</u> A. We have four resistors A, B, C and D of resistance $3\ \Omega$, $6\ \Omega$, $9\ \Omega$ and $12\ \Omega$ respectively. Find out the lowest resistance which can be obtained by combining these four resistors. OR B. You are given 2 fuse wires A and B with current ratings 2A and 5A respectively. Justify with reason which of the two would you use with a 1000W, 220V room heater?	2												

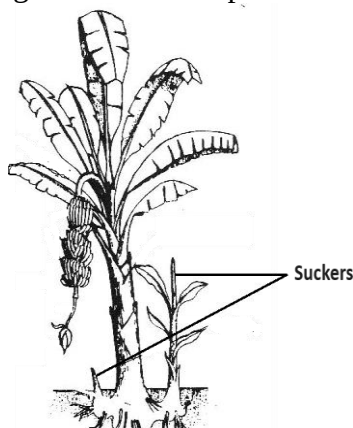
Section-C
Question No. 27 to 33 are short answer questions

27	A teacher asks her students to identify a metal, M. She gives them the following clues to help them. * Its oxide reacts with both HCl and NaOH. * It does not react with hot or cold water but reacts with steam. * It can be extracted by electrolysis of its ore. A. Identify the metal. B. Write the chemical equations for the reaction of the metal with HCl and NaOH respectively. C. What would happen if the metal is reacted with iron oxide?	3
28	<u>Attempt either option A or B.</u> A. A chemical compound 'X' is used in the soap and glass industry. It is prepared from brine. (i) Write the chemical name, common name and chemical formula of 'X'. (ii) Write the equation involved in its preparation. (iii) What happens when it is treated with water containing Ca and Mg salts? OR B. Comment on the following statements: (i) Bee sting is treated with baking soda paste whereas wasp sting is treated with dilute vinegar. (ii) Farmers treat soil with quicklime when tilling. (iii) Ancient sculptures and marble structures are conserved by treating them with certain chemicals.	3
29	What are the methods used by plants to get rid of excretory products?	3
30	Name the plant Mendel used for his experiment. What type of progeny was obtained by Mendel in F ₁ and F ₂ generations when he crossed the tall and short plants? Write the phenotypic and genotypic ratio he obtained in F ₂ generation plants.	3
31	The far point and the near point refer to the visibility of objects close by and far away from the human eye respectively. These are the maximum and minimum distances at which an object is clearly visible to a person.  A. The near point and the far point are determined with regards to the function of which part of the eye? B. Why a far sighted person needs a convex lens to correct his vision? C. A person is advised to wear spectacles with concave lenses. What type of defect of vision is he suffering from?	3
32	A. Write the function of voltmeter and ammeter. B. Should the resistance of a voltmeter be low or high? Give reason.	3
33	A helical coil whose length is greater than its diameter is connected to a battery as shown below.  A. How does the magnetic field at point P compare with the magnetic field at point Q? Justify your answer. B. State one way in which the strength of the magnetic field inside a current carrying helical coil can be changed?	3

Section-D Question No. 34 to 36 are long answer questions.
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5

A. The image below shows a banana plant which is growing with the help of suckers. These suckers are small plant stem outgrowths which can be separated from the main plant and planted separately and they will grow into a new plant subsequently.



- Give the name and type of reproduction that is shown in the image above.
- List two advantages the farmer will have on using the type of reproduction mentioned above.
- The above plant produces male flowers. Explain how this plant will be involved in the process of pollination.
- Why is the offspring of this banana plant not absolutely identical to its parent plant?

OR

B. A developing fetus is connected to the placenta by means of umbilical cord. The Umbilical vein and artery run inside the umbilical cord.

- (i) Name two substance that moves through the blood vessels.
- (ii) If the placenta has less villi how will it affect the baby's growth?
- (iii) Name the region where the embryo develops inside the female body. Explain how this region is adapted for nourishing the baby.
- (iv) Some of the fetal cells fall off into the amniotic fluid and can be collected by careful procedure. The cells were screened and found to contain XY chromosome.
 - a) What is the sex of the fetus?
 - b) How is this prenatal sex determination misused?

A. (i) Rehmat classified the reaction between Methane and Chlorine in presence of sunlight as a substitution reaction. Support Rehmat's view with suitable justification and illustrate the reaction with the help of a balanced chemical equation.

(ii) Chlorine gas was prepared using electrolysis of brine solution. Write the chemical equation to represent the change. Identify the other products formed in the process and give one application of each.

OR

B. Raina while doing certain reactions observed that heating of substance 'X' with vinegar like smell with a substance 'Y' (which is used as an industrial solvent) in presence of conc. Sulphuric acid on a water bath gives a sweet-smelling liquid 'Z' having molecular formula $C_4H_8O_2$. When heated with caustic soda (NaOH), 'Z' gives back the sodium salt of and the compound 'Y'.

Identify 'X', 'Y', and 'Z'. Illustrate the changes with the help of suitable chemical equations.

5

A. Two identical resistors, each of resistance $15\ \Omega$, are connected in (i) series and (ii) parallel, in turn to a battery of $6\ \text{V}$. Calculate the ratio of the power consumed in the combination of resistors in each case.

OR

B. (i) State Joules law of heating and write its mathematical expression.

(ii) Two resistors of resistances 2Ω and 4Ω are connected in

b) parallel

with a battery of given potential difference. Compute the ratio of total quantity of heat produced in the combination in the two cases if the total voltage and time are kept the same for both.

Section – E

Question No. 37 to 39 are case-based/data -based questions.

37	Read the following passage and answer the questions that follows. Metals are required for a variety of purposes. For this we need their extraction from their ores. Ores mined from the earth are usually contaminated with many impurities which must be removed prior to the extraction of metals. The extraction of pure metal involves the following steps: • Concentration of ore • Extraction of the metal from the concentrated ore • Refining of the metal A. Name an ore of Mercury and state the form in which Mercury is present in it. B. What happens to Zinc carbonate when it is heated strongly in a limited supply of air? Attempt either subpart C or D. C. The reaction of a metal A with Fe_2O_3 is highly exothermic and is used to join railway track. Identify the metal A and name the reaction, also write the chemical equation of the reaction. OR D. We cannot use carbon to obtain sodium from sodium oxide. Why? State the reactions taking place at cathode and anode during electrolytic reduction of sodium chloride.	4
38	Azim Taraporewala was a traveller and science enthusiast. During one of his travels he found himself on the edge of an island without any mode of communication. As he had read in many stories, he thought he would light a fire on the beach and travelling boats or ships could see that fire and come to give him a ride. He had run out of lighters and match-sticks but had a reading glass. Being a science enthusiast he knew some tricks and used that lens and a scrap of paper to light a fire, with the help of scorching rays from the sun. A. Which lens can be used by Azim to create the fire? B. What property of the lens helps Azim to create the fire? Attempt either option C or D. C. List two more uses of this kind of lens. OR D. Explain with reason the condition under which the lens can form both real as well as virtual images.	4
39	Some plants like the pea plant climb up other plants or fences by means of tendrils. These tendrils are sensitive to touch. When they come in contact with any support, the part of the tendril in contact with the object does not grow as rapidly as the part of the tendril away from the object. This causes the tendril to circle around the object and thus cling to it. More commonly, plants respond to stimuli slowly by growing in a particular direction. Because this growth is directional, it appears as if the plant is moving. A. How many type of tropism are shown by plants? Name them. Attempt either option A or B. B. The touch me not plant is an example of which tropism? Name the plant hormone which inhibits growth? C. Give one example of chemotropism? D. Name the plants hormone which promotes cell division?	4